

dhIA (P22643, XANAU) — Epoxide Hydrolase Activity (GO:0004301)

Hypothesis Review

Focus type: function_assignment **Seed hypothesis:** dhIA has epoxide hydrolase activity (GO:0004301). **Decision context:** GO:0004301 (epoxide hydrolase activity), Evidence IEA, original reference GO_REF:0000118. **Source:** `genes/XANAU/dhIA/dhIA-ai-review.yaml` → `existing_annotations[2].function_hypothesis`

Executive Judgment

Verdict: Over-annotated (refuted as a direct function).

The gene product P22643/DHLA_XANAU is the archetypal **haloalkane dehalogenase** (EC 3.8.1.5) from *Xanthobacter autotrophicus* GJ10. Its experimentally established, curated molecular function is hydrolytic cleavage of carbon–halogen bonds (GO:0018786, haloalkane dehalogenase activity). The `GO:0004301` epoxide hydrolase annotation is an **IEA:TreeGrafter** assignment (consistent with the supplied GO_REF:0000118) that arises purely from the shared **α/β -hydrolase fold** and the InterPro "**Epoxide hydrolase-like**" **signature (IPR000639 / PRINTS EPOXHYDRLASE)**. Epoxide hydrolases (EC 3.3.2.x) and haloalkane dehalogenases are homologous, use the same fold and an Asp–His–Asp/Glu triad with a covalent enzyme intermediate, but are **distinct enzymatic activities** — interconverting them requires deliberate active-site engineering. No primary literature or curated UniProt catalytic-activity statement reports epoxide hydrolysis by DhIA.

Most important caveat: The annotation is not biologically absurd (the fold genuinely is epoxide-hydrolase-like, and promiscuous low-level activity on some epoxides cannot be formally excluded without an assay), but it is **less precise than existing experimental knowledge** and should be treated as a non-core, computationally derived term that a curator would normally not retain for this well-characterized enzyme.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
UniProt P22643 (DHLA_XANAU), curated record	Review/ database	Refutes (direct function is dehalogenase)	What is DhIA's curated activity?	Rec. name "Haloalkane dehalogenase", EC 3.8.1.5; two CATALYTIC ACTIVITY statements are C-halogen hydrolysis only; no epoxide reaction listed	Swiss-Prot curated entry X. autotrophic
UniProt P22643 GO annotations	Database	Qualifies (identifies source)	Where does GO:0004301 come from?	GO:0004301 = IEA:TreeGrafter ; GO:0018786 (dehalogenase) = IEA:UniRule; GO:0019260 (1,2-DCE catabolism) = UniPathway	Automated pipeline
UniProt P22643 domain cross-refs	Structural/ evolutionary, computational	Qualifies (explains over-transfer)	Why is an EH term applied?	Matches IPR000639 "Epox_hydrolase-like", PRINTS PR00412 EPOXHYDRLASE, under α/β -hydrolase fold (SSF53474, PF00561, IPR029058); PANTHER PTHR42977 "HYDROLASE-RELATED"	Sequence/ domain signatures

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
Blazic, Gautier, Norberg, Widersten 2024, P 38828992	Direct assay / protein engineering	Refutes / Qualifies	Are EH and DhIA the same activity?	"Epoxide hydrolase StEH1 ... is similar in overall structural fold and catalytic mechanism to haloalkane dehalogenase DhIA"; alkyl- halide hydrolase activity had to be engineered into an EH scaffold	Potato StEH1 vs DhIA, in vitro
Argiriadi, Morisseau, Hammock, Christianson 1999, P 10485878	Structural / evolutionary	Qualifies	Fold relationship EH↔dehalogenase	EH catalytic domain "is similar to that of haloalkane dehalogenase and shares the α / β hydrolase fold"; proposes shared evolutionary sequence	Murine cytosolic EH crystal structure
Kokkonen et al. 2018 (P 29858243); Bosma et al. 2003 (P 12834356); Olsson & Warshel 2004 (P 15548014)	Direct assay / mechanism	Refutes	What reaction does DhIA catalyze?	DhIA/DhAA catalyze hydrolysis of carbon-halogen bonds via Asp nucleophile + His/Asp; covalent alkyl-enzyme intermediate; SN2 on haloalkane	Enzymology/ kinetics/QM- MM

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
UniProt REST family audit (this report, Iter 2)	Computational / database	Qualifies / Refutes	Is GO:0004301 a conserved family function?	33/33 reviewed EC 3.8.1.5 enzymes carry GO:0018786; only 13/33 (39%) carry GO:0004301, all IEA; absent from DhaA subgroup & LinB; no experimental EH annotation anywhere	33 Swiss-Pro haloalkane dehalogenase

QuickGO Annotation Inventory for P22643 (computed provenance, Iteration 3)

Complete set of GO annotations on P22643 (EBI QuickGO API) — **all four are IEA**:

GO ID	Term	Evidence	Reference	Assigned by
GO:0003824	catalytic activity	IEA	GO_REF:0000002	InterPro
GO:0004301	epoxide hydrolase activity	IEA	GO_REF:0000118	TreeGrafter
GO:0018786	haloalkane dehalogenase activity	IEA	GO_REF:0000120	UniProt (UniRule)
GO:0019260	1,2-dichloroethane catabolic process	IEA	GO_REF:0000041	UniProt

- **Confirms directly that GO_REF:0000118 = the TreeGrafter pipeline** (QuickGO `assignedBy` field).
- Despite DhIA being one of the most experimentally characterized α/β -hydrolases, **no experimental (EXP/IDA/IMP) GO annotation exists** — every term is electronic. The

dehalogenase term is nonetheless backed by UniRule + the curated EC 3.8.1.5 reaction and decades of primary enzymology; the epoxide hydrolase term is TreeGrafter-only and has no corresponding assay.

Family-Wide Annotation Audit (computed provenance, Iteration 2)

UniProt REST query of **all 33 reviewed Swiss-Prot EC 3.8.1.5 haloalkane dehalogenases**, checking presence of each GO term:

GO term	Label	Count carrying it	Fraction	Evidence code
GO:0018786	haloalkane dehalogenase activity	33 / 33	100%	all IEA
GO:0004301	epoxide hydrolase activity	13 / 33	39%	all IEA (TreeGrafter)

- The core dehalogenase term is **universal**; the epoxide hydrolase term is **patchy**.
- GO:0004301 is present in DHLA_XANAU/DHLA_XANFL and the **Dhma** subgroup (Mycobacterium/Caulobacter) but **absent from the entire DhaA subgroup** (e.g., DHAA_RHORH/P0A3G2, DHAA_MYCTU/P9WMR9) and from **LinB** (D4Z2G1).
- **Zero** of the 33 enzymes carry an experimental (non-IEA) epoxide hydrolase annotation.

Interpretation: a genuinely conserved catalytic activity would annotate uniformly; the 39% patchiness tracks TreeGrafter/PANTHER subtree placement, i.e. classic homology-driven over-propagation of the sibling fold activity — not evidence of real epoxide hydrolysis.

GO Curation Implications

- **Lead (requires curator verification):** Treat **GO:0004301 (epoxide hydrolase activity, IEA/TreeGrafter)** as a **non-core, over-propagated computational term**. Recommended

action: **remove or demote** (do not retain as a representative function) for P22643, because a more specific, experimentally grounded MF term already exists.

- **Retain as the primary MF term: GO:0018786 (haloalkane dehalogenase activity)** — matches the curated EC 3.8.1.5 and all primary enzymology.
 - **Retain BP: GO:0019260 (1,2-dichloroethane catabolic process)** — pathway context.
 - The epoxide-hydrolase term is **too broad/incorrect in specificity** for this gene: it is an MF sibling activity on the same fold, not DhIA's demonstrated activity. It should not be generalized or made more specific — it should be flagged as homology carry-over.
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Mechanistic Scope

- **Immediate molecular activity tested:** hydrolysis of an epoxide C–O bond (epoxide hydrolase).
 - **What DhIA actually does (direct):** nucleophilic Asp124 performs SN2 attack on the terminal carbon of a haloalkane, displacing halide and forming a covalent **alkyl-enzyme ester**, which is then hydrolyzed by a water activated by the His289–Asp260 pair — yielding a primary alcohol, halide, and H⁺.
 - **Relationship to epoxide hydrolysis:** EH uses the *same fold and an analogous Asp–His–acid triad* but opens an epoxide ring (covalent alkoxy/ester intermediate). The mechanistic similarity is the reason for the mis-transfer; it is **not** evidence that DhIA performs epoxide hydrolysis.
 - Epoxide hydrolysis would be a **separate catalytic function**, not a downstream phenotype — so the question is purely whether DhIA has this activity, and the record indicates it does not (as a demonstrated function).
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Conflicts and Alternatives

- **Database carry-over / paralog-family confusion (most likely explanation):** TreeGrafter grafts the sequence onto a PANTHER family (PTHR42977 "HYDROLASE-RELATED") whose members include epoxide hydrolases; the InterPro "Epoxide hydrolase-like" signature (IPR000639) and PRINTS EPOXHYDRLASE then license GO:0004301. This is frequency/fold bias, not organism-specific biology.

- **Promiscuity caveat (cannot be fully excluded):** α/β -hydrolase enzymes are catalytically promiscuous; low-level, in-vitro-only epoxide hydrolysis by DhIA has not been reported but also has not been exhaustively assayed. Even if present, promiscuous activity would be non-core and should not be annotated without an experimental reference.
 - **No conflicting primary evidence** supports epoxide hydrolase activity as a physiological function.
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Knowledge Gaps

1. **Has DhIA ever been assayed against epoxides (e.g., epichlorohydrin, styrene oxide, 1,2-epoxyalkanes)?** Checked UniProt catalytic-activity statements and primary enzymology abstracts — none report epoxide turnover. Matters because a single positive assay would convert "over-annotation" into "non-core promiscuous activity." Resolver: direct spectrophotometric/GC epoxide-hydrolase assay on purified DhIA.
 2. **Exact PANTHER subfamily/tree node driving the TreeGrafter call.** Checked UniProt cross-refs (PTHR42977:SF3). Matters because knowing the grafted family clarifies whether the whole subfamily is mis-annotated. Resolver: inspect the PANTHER tree and GO_REF:0000118 propagation rules.
 3. **Whether sibling haloalkane dehalogenases (DhaA, LinB, DbjA) carry the same GO:0004301 IEA.** Not yet queried. Matters for systematic correction across the family. Resolver: batch UniProt/QuickGO query.
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Discriminating Tests

- **Direct epoxide hydrolase assay** on purified recombinant DhIA (e.g., styrene oxide, epichlorohydrin, 1,2-epoxyhexane) measuring diol formation vs. a known EH positive control and a buffer blank — the single most decisive experiment.
- **Active-site logic check:** DhIA's halide-binding pocket (Trp125/Trp175) is tuned to stabilize a departing halide, not an alkoxide/diol; structural comparison with StEH1/human EH active sites predicts poor epoxide turnover. A docking/structure comparison is a cheap computational discriminator.

- **Family-wide annotation audit:** QuickGO/UniProt query of all PTHR42977 members and all EC 3.8.1.5 enzymes to quantify how broadly GO:0004301 is TreeGrafter-propagated (frequency-bias test).

Curation Leads (require curator verification)

- **Candidate action:** Do **not** accept GO:0004301 as a represented molecular function for dhla; mark the IEA/TreeGrafter epoxide hydrolase term as **over-annotated** / **non-core** and prefer GO:0018786 (haloalkane dehalogenase activity) + GO:0019260 (1,2-dichloroethane catabolic process).

- **Candidate reference + snippet to verify**

(P 38828992):

"Epoxide hydrolase StEH1, from potato, is similar in overall structural fold and catalytic mechanism to haloalkane dehalogenase Dhla from" — use to document that fold/mechanism similarity (not a real shared activity) explains the term; the two activities had to be engineered apart.

- **Candidate reference + snippet**

(P 10485878):

"this domain is similar to that of haloalkane dehalogenase and shares the alpha/beta hydrolase fold" — supports homology-driven mis-transfer.

- **Database provenance to cite:** UniProt P22643 shows GO:0004301 = IEA:TreeGrafter (= GO_REF:0000118), while the curated reaction statements are strictly EC 3.8.1.5 dehalogenase.
 - **Suggested curator question:** Is there *any* experimental reference showing epoxide turnover by Dhla? If none, remove/demote GO:0004301.
 - **Suggested experiment:** Spectrophotometric epoxide hydrolase assay on purified Dhla with EH-positive control.
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Limitations

- This review relied on literature and the public UniProt/InterPro records; no wet-lab assay was run.
 - Absence of a reported epoxide-hydrolase assay is evidence of absence of *demonstrated* activity, not proof that trace promiscuous activity is impossible.
 - GO_REF:0000118 ↔ TreeGrafter equivalence was inferred from the UniProt evidence string "IEA:TreeGrafter"; curators should confirm the exact GO_REF mapping in their pipeline.
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